

Venkateswaran Subramaniam

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EDUCATION

BITS PILANI

Pilani, Rajasthan, India

MSC IN BIOLOGICAL SCIENCES

Graduated July 2024

CGPA : 8.45/10

First Division

BE IN ELECTRONICS ENGINEERING

Graduated July 2024

CGPA : 8.45/10

First Division

COURSEWORK

SOFTWARE

Object Oriented Programming

Data Structures

Programming in Python

C Programming

HARDWARE

Digital VLSI Design

Analog Circuit Design

Embedded Systems Design

Digital Signal Processing

Microcontroller Design/Interfacing

Electronic Circuit Analysis & Design

SELF TAUGHT

Computer Architecture & Verilog

Embedded drivers development

FPGA programming

SKILLS

PROGRAMMING

Experienced:

Python • C • Verilog • SystemVerilog

Familiar:

UVM • Perl • tcl • Xilinx vivado

TOOLS/APPLICATIONS

Visual Studio • Eclipse • STM32CubeIDE

• MATLAB • Code Composer Studio •

Altium Designer • Keil µvision 5

OPERATING SYSTEMS

Windows • Linux • UNIX

EXPERIENCE

CENIZAS LABS | HARDWARE ENGINEER

July. 2024 – present | Chennai, India

- R&D and Testing of a low latency (100µs) voltage stabilizer for AC applications.
- Led 30+ tests to evaluate buck/boost performance under varying conditions (MOSFETs, diodes, LC size, gate drivers, op-amp drivers, RMS ICs).
- Validated design changes using LTSPICE and ModelSim, achieving 95% correlation between simulation and hardware results.
- Designed and routed a 4-layer PCB for the TAKTUS balance board, reducing signal integrity issues related to the RF antenna module.
- Currently developing a firmware abstraction layer for STM32 and TI microcontrollers, including an automated sensor driver development and integration tool that aims to reduce development time by 50%.

NVIDIA GRAPHICS | MEMORY SUBSYSTEM VERIFICATION INTERN

Feb 2024 – July 2024 | Bangalore, India

- Implemented low weight coverage runs for the UCF-SoC TB due to which major bugs are caught early, reducing the verification effort by as much as 50%.
- Developed functional coverage automation scripts in python that could generate the coverage class when given a simplified yaml file as an input.
- Monitored coverage reports, reporting any unexpected behaviors in memory transactions and coverage holes related to packet src, dest, transaction type etc.
- Maintained flow related scripts (Python, Perl, tcl) related to the above tasks.

DESIGN EXPERIENCE

FPGA-BASED COMMUNICATION PROTOCOL DRIVERS

Personal Project, Fall 2024

- Successfully designed and implemented drivers for SPI, UART, and I2C protocols on a Xilinx Spartan-7 Cmod S7, enabling seamless data exchange between peripherals and the FPGA.
- Developed and optimized Verilog/VHDL code to ensure efficient resource utilization and low- latency performance in a constrained FPGA environment.
- Conducted testing and validation using Xilinx Vivado and ModelSim, ensuring reliable functionality through simulation and debugging.

SMART PARKING SYSTEM

Course Project, Fall 2022

- Developed an automated parking system using STM32F4 Microcontroller, MRFC522 RFID Module, and IR sensors to detect vehicle presence and display real-time slot availability on an LCD screen.
- Improved parking efficiency by enabling seamless vehicle identification, reducing driver search time, and providing real-time updates on available slots.

EXTRACURRICULARS AND HOBBIES

Chess • Public Speaking - Toastmasters • Badminton • Table Tennis • Game Development • Learning Languages